TASK-4 SIMPLE IOT : ESP32 WEB DASHBOARD

1. Project Overview

The Internet of Things (IoT) is a technology that enables physical devices to communicate and exchange data over the internet. One of the most common IoT applications is remote environmental monitoring using sensors and web-based dashboards. This project demonstrates an IoT-based temperature and humidity monitoring system using an ESP32 microcontroller and a DHT22 sensor. The ESP32 reads environmental data from the sensor and hosts a web server that displays the measured values on a web dashboard. Users can monitor the sensor readings through a web browser connected to the same network. This project introduces the concepts of wireless communication, web server development, sensor interfacing, and real-time data monitoring using IoT technology.

2. Aim / Objective

The objective of this project is to develop an IoT-based monitoring system using an ESP32 microcontroller that measures temperature and humidity using a DHT22 sensor and displays the measured values on a web dashboard accessible through a web browser.

3. Components Required

- ESP32 Development Board
- DHT22 Temperature and Humidity Sensor
- LED
- 220 Ω Resistor
- Breadboard
- Jumper Wires

4. Software Required

- Arduino IDE
- Wokwi Simulator
- ESP32 Board Package
- Web Browser (Google Chrome, Microsoft Edge, etc.)

5. Theory

The Internet of Things (IoT) is a network of interconnected devices capable of collecting, processing, and exchanging data over the internet. IoT systems are widely used in smart homes, healthcare, industrial automation, agriculture, and environmental monitoring.

In this project, the ESP32 microcontroller serves as the central processing unit and Wi-Fi communication device. The ESP32 is equipped with built-in Wi-Fi capabilities, allowing it to host a web server without requiring any additional communication modules. The DHT22 sensor is used to measure both temperature and relative humidity with good accuracy and reliability. The sensor sends digital data to the ESP32, which processes the readings and updates the web dashboard accordingly.

The ESP32 creates a lightweight web server that can be accessed through a web browser using its IP address. The web page displays the current temperature and humidity values in an organized and user-friendly format. In addition, an LED can be controlled through the dashboard, demonstrating remote device control using IoT technology.

This project combines sensor interfacing, wireless communication, embedded programming, and web technologies to create a simple yet effective IoT monitoring system. It provides a practical understanding of how embedded devices communicate over networks and how real-time sensor data can be visualized through web-based interfaces.

6. Working Principle

The ESP32 connects to a Wi-Fi network and initializes the DHT22 temperature and humidity sensor. The sensor continuously measures the surrounding temperature and humidity and sends the digital readings to the ESP32. The ESP32 processes the received data and hosts a web server that generates a web page displaying the latest sensor values. Whenever a user opens the web page through a browser, the current temperature and humidity readings are displayed in real time. If LED control is included, users can also switch the LED ON or OFF through the web dashboard. The entire process repeats continuously, enabling real-time environmental monitoring over the network.

Source code:

```
#include <WiFi.h>
#include <WebServer.h>
#include <DHTesp.h>
const char* ssid = "Wokwi-GUEST";
const char* password = "";
WebServer server(80);
DHTesp dht;
const int ledPin = 2;
bool ledState = false;

void handleRoot() {
  TempAndHumidity data = dht.getTempAndHumidity();
  String html = R"rawliteral(
```

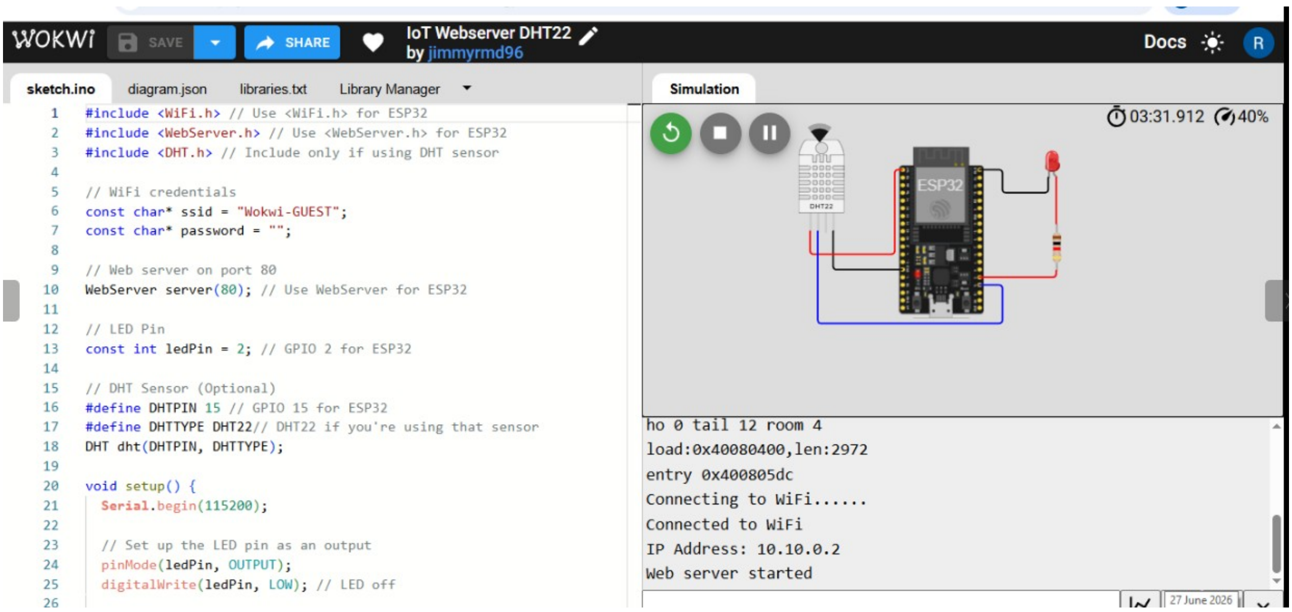
```
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>ESP32 IoT Dashboard</title>
<style>
body{
background:#f2f5f9;
font-family:Arial;
text-align:center;
margin:0;
padding:20px;
}
h1{
color:#1565c0;
}
.container{
display:flex;
justify-content:center;
gap:20px;
flex-wrap:wrap;
margin-top:30px;
}
.card{
background:white;
width:250px;
padding:20px;
```

```
border-radius:12px;
box-shadow:0 4px 10px rgba(0,0,0,0.2);
}
.value{
font-size:38px;
font-weight:bold;
color:#1565c0;
}
button{
background:#1976d2;
color:white;
border:none;
padding:12px 30px;
border-radius:8px;
font-size:18px;
cursor:pointer;
margin-top:20px;
}
button:hover{
background:#0d47a1;
}
</style>
</head>
<body>
<h1>ESP32 IoT Dashboard</h1>
<div class="container">
<div class="card">
<h2>🌡️ Temperature</h2>
```

```
<div class="value">
  html += String(data.temperature, 1);
  html += " °C</div>";
  html += "</div>";
  html += "<div class='card'>";
  html += "<h2>💧 Humidity</h2>";
  html += "<div class='value'>";
  html += String(data.humidity, 1);
  html += " %</div>";
  html += "</div>";
  html += "</div>";
  html += "<div style='margin-top:30px;'>";
  html += "<h2>💡 LED Control</h2>";
  if (ledState) {
    html += "<p><b>LED Status : ON</b></p>";
    html += "<a href='/off'><button>Turn OFF</button></a>";

  }
else {
    html += "<p><b>LED Status : OFF</b></p>";
    html += "<a href='/on'><button>Turn ON</button></a>";

  }
  html += "</div>";
  html += "</body></html>";
  )rawliteral";
  server.send(200, "text/html", html);
}
```

```
void handleOn() {
  ledState = true;
  digitalWrite(ledPin, HIGH);
  server.sendHeader("Location", "/");
  server.send(303);
}
void handleOff() {

  ledState = false;
  digitalWrite(ledPin, LOW);
  server.sendHeader("Location", "/");
  server.send(303);

}
void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);
  dht.setup(15, DHTesp::DHT22);
  WiFi.begin(ssid, password);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print(".");

  }

  Serial.println();
  Serial.println("Connected to WiFi");
```

```
  Serial.print("IP Address: ");
  Serial.println(WiFi.localIP());
  server.on("/", handleRoot);
  server.on("/on", handleOn);
  server.on("/off", handleOff);
  server.begin();
  Serial.println("Web server started");
}
void loop() {
  server.handleClient();
  delay(10);
}
```



7. Algorithm

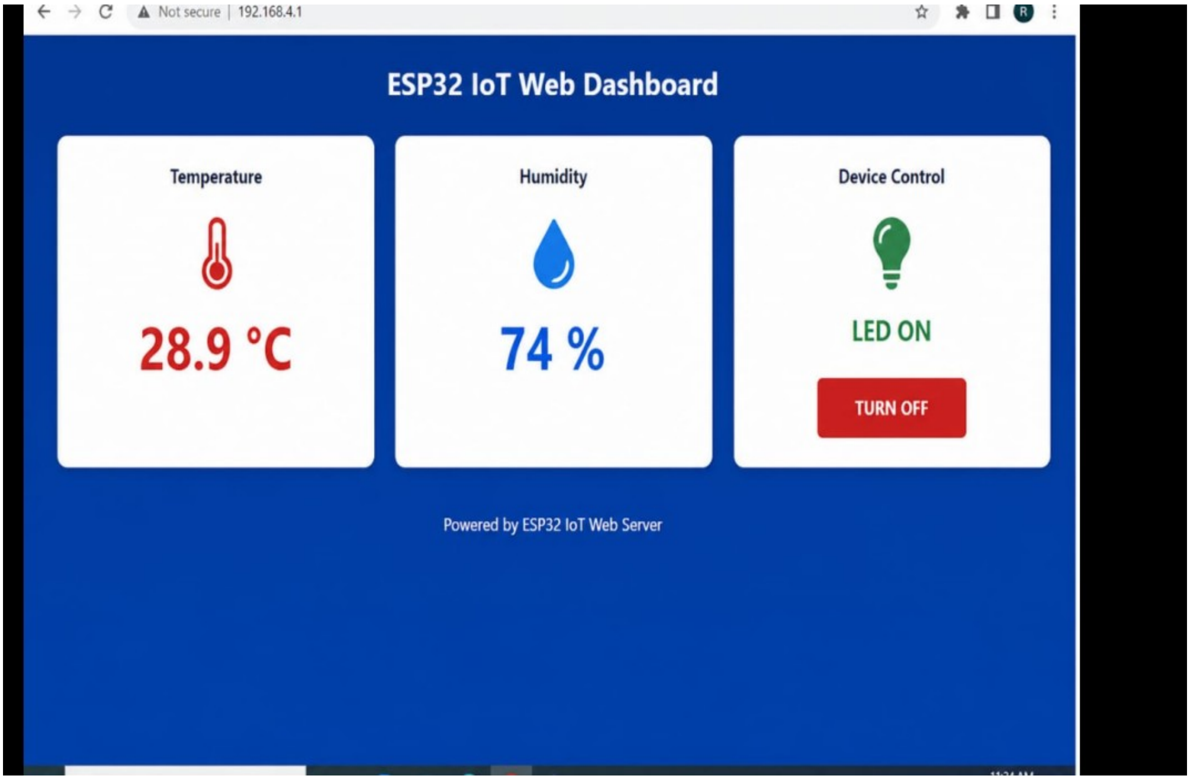
1. Start the program.
2. Initialize the ESP32, DHT22 sensor, and Wi-Fi connection.
3. Connect the ESP32 to the available Wi-Fi network.
4. Start the embedded web server.
5. Read temperature and humidity values from the DHT22 sensor.
6. Update the web page with the latest sensor readings.

7. Monitor user requests from the web browser.
8. If LED control is enabled, process the ON/OFF commands.
9. Repeat the process continuously to provide real-time monitoring.

8. How to Run

1. Open the project in the Wokwi simulator.
2. Verify the circuit connections and ESP32 program.
3. Start the simulation.
4. Wait for the ESP32 to connect to the simulated Wi-Fi network.
5. Open the generated web dashboard in the browser.
6. Observe the temperature and humidity readings displayed on the dashboard.
7. If available, use the dashboard controls to switch the LED ON or OFF.

Output:



9. Applications

- Smart home automation
- Environmental monitoring systems
- Weather monitoring stations
- Industrial process monitoring
- Smart agriculture

- Greenhouse monitoring
- Healthcare monitoring systems
- IoT-based remote monitoring applications

10. Skills Gained

- ESP32 programming using Arduino IDE.
- IoT application development.
- Wi-Fi communication and networking.
- Web server implementation on ESP32.
- DHT22 sensor interfacing.
- Real-time sensor data monitoring.
- Embedded web development.
- Circuit simulation using Wokwi.
- Debugging IoT applications using the Serial Monitor.

11. Conclusion

The ESP32 IoT Web Dashboard project successfully demonstrates the implementation of a wireless environmental monitoring system using IoT technology. The ESP32 collects temperature and humidity data from the DHT22 sensor and displays the information on a web dashboard accessible through a standard web browser. The project provides practical experience in embedded programming, wireless communication, sensor interfacing, and web server development. It serves as an excellent foundation for building advanced IoT applications involving cloud connectivity, remote monitoring, and smart automation systems.

Internship :

Completed as part of Embedded Systems Internship – Intern spark